Empirically: $R_a \propto \exp(h \cdot B_a)$

$R_t \propto \exp(h \cdot B_t)$

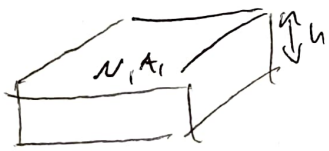
$$R_a = \frac{1}{\sigma_a} \cdot \frac{h}{L \cdot W} \rightarrow \sigma_a = \frac{h}{R_a (LW)}$$

$$R_t = \frac{1}{\sigma_t} \cdot \frac{L}{h \cdot W} \rightarrow \sigma_t = \frac{L}{R_t \, hW}$$

Ansatz: $R_a = N^{\alpha} A^{\beta} \cdot f(h \cdot N^{\delta} A^{\varepsilon})$

$(A = LW)$

$\downarrow I$



$R_a' = N'^{\alpha} A'^{\beta} f(h' N'^{\delta} A'^{\varepsilon})$

$\Rightarrow$ $R' \rightarrow 2R$

$h' \rightarrow 2h$

$A' \rightarrow A$

$N' \rightarrow 2N$

$R_a' = 2^{\alpha} N^{\alpha} A^{\beta} \, f(2h \cdot 2^{\delta} N^{\delta} A^{\varepsilon})$

$\rightarrow \delta = -1$

$\alpha = 1$

axial cont'd:

$R' \to \frac{1}{2} R$

$h' \to h$

$N' \to 2N$

$A' \to 2A$

$$R_a = \underbrace{2^{\alpha} N^{\alpha} 2^{\beta} A^{\beta}}\, f\left(h \cdot 2^{\delta} N^{\delta} 2^{\varepsilon} A\right)$$

$\delta = -1$ from before

$\Rightarrow \varepsilon = 1$ so that $\delta + \varepsilon = 0$

$\alpha + \beta = -1$

$\alpha = 1$

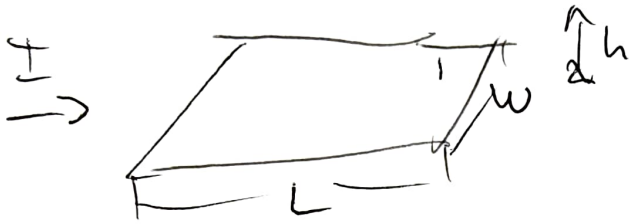
$\beta = -1 - 1 = -2$

$$\boxed{R_a = \frac{N}{A^2} f\left(\frac{hA}{N}\right)}$$

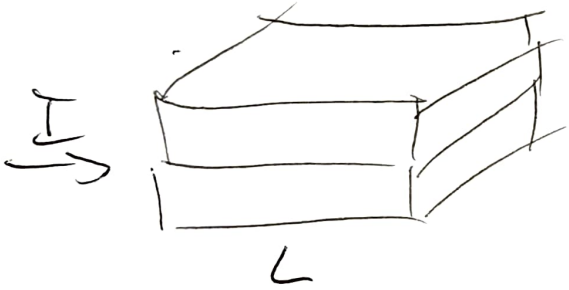
prefactors?

transverse: $R_t \propto \exp(h \cdot B_t)$

~~Ansatz:~~



Ansatz: $R_t = N^{\alpha} L^{\beta} w^{\gamma} f(h \cdot L^{\delta} w^{\varepsilon} N^{\sigma})$



$N' \to 2N$

$L' \to L$

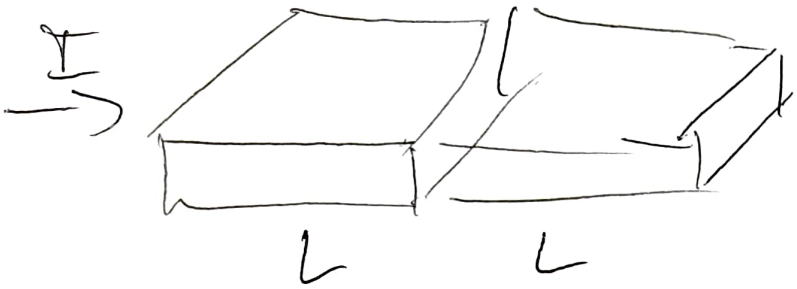
$w' \to w$

$h' \to 2h$

$R' \to \frac{1}{2} R$

$2^{\alpha} \quad f(2h \cdot 2^{\sigma})$

$\Rightarrow \alpha = -1 \quad , \sigma = -1$



$R' \to 2R$

$L' \to 2L$

$W' \to W$

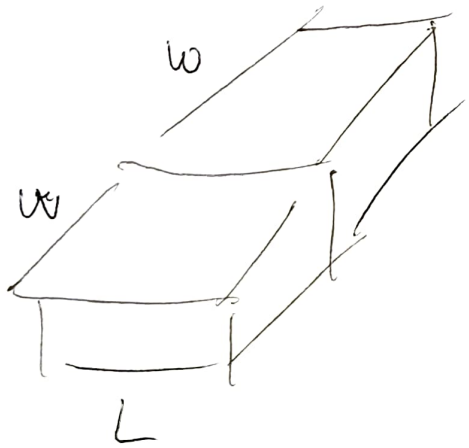
$h' \to h$

$N' \to 2N$

$$2^{\overset{\alpha = -1}{\alpha}}\, 2^{\beta}\, f\left(2^{\delta}\, 2^{\overset{=-1}{\sigma}}\right)$$

$\alpha + \beta = 1 \Rightarrow \beta = 2$

$\delta + \sigma = 0 \to \delta = 1$



$R' \to \frac{1}{2} R$

$N' \to 2N$

$h' \to h$

$L' \to L$

$W' \to 2W$

$$n \sim 2^{\overset{-1}{\alpha}}\, 2^{\gamma}\, f\left(2^{\varepsilon}\, 2^{\overset{-1}{\sigma}}\right)$$

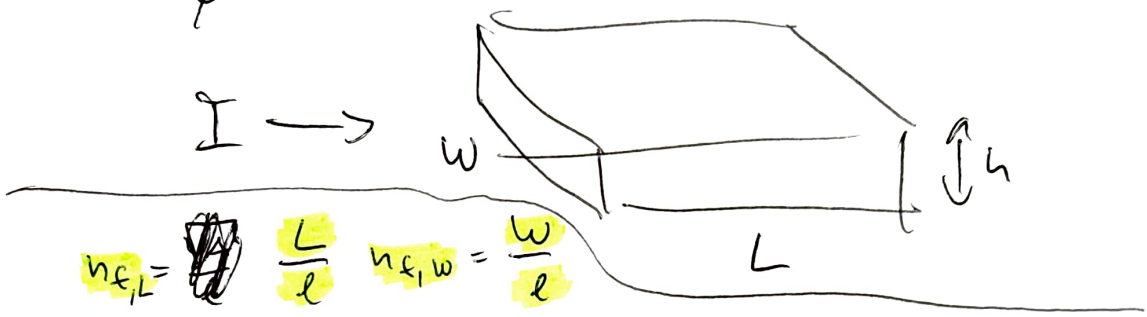
$\gamma = 0$

$\varepsilon + \sigma = 0 \to \varepsilon = 1$

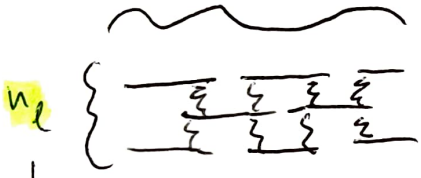
$R_t = \frac{L^2}{N} \cdot \exp\left(\frac{LWh}{N}\right)$ & Prefactors

$$R_a = \frac{N}{A^2}\exp\left(\frac{Ah}{N}\right) \qquad R_T = \frac{L^2}{N}\exp\left(\frac{LWh}{N}\right)$$

$$\sigma_a = \frac{V_f}{\phi}e^{-(V_f/\phi)} \qquad \sigma_T = \frac{\phi}{V_f}e^{-(V_f/\phi)}$$



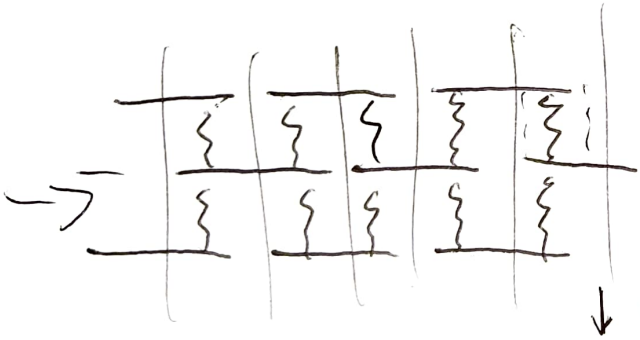
$n_{f,L} = \frac{L}{\ell}$ $\quad n_{f,w} = \frac{W}{\ell}$



Assume Brickleyard structure

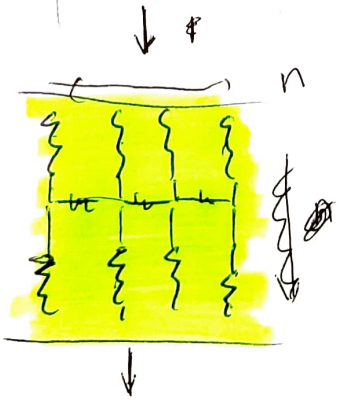
↳ number of layers: $n_\ell = \frac{N\ell^2}{A}$

$V_f = \ell^2 \cdot t$

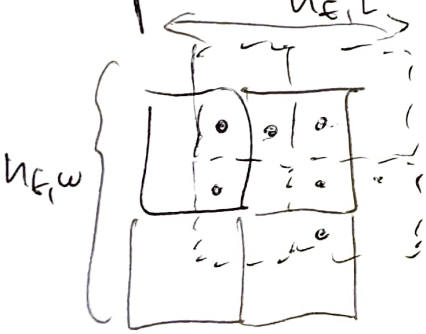


axial: No of resistors per channel n_ℓ

No. of channels is $n_{f,L} \cdot n_{f,w}$



top view



$$\frac{1}{R_{tot}} = \overbrace{\frac{1}{R} + \frac{1}{R} + \dots}^{n}$$

$$R_{total} = \frac{R}{n}$$

axial: each channel has n_ℓ resistors $R(\phi)$

$$R_{total} = \frac{n_\ell R(\phi)}{n_{\ell,L} \cdot n_{L,W}}$$

— no. of Elements
no. of channels in each layer.

$$R_{total} = \frac{N\ell^2}{L \cdot W} \frac{\ell}{L} \frac{1}{W} R(\phi) = \frac{N}{A^2} \ell^4 R(\phi) \rightarrow \text{axial}$$

transverse

no. of layers is $n_{f,L}$

$$R_{total} = \frac{n_{f,L} \cdot R(\phi)}{n_\ell \cdot n_{f,W}}$$

$$= \frac{L \cdot L \cdot W}{\ell N \ell^2 W} = \frac{L^2}{N\ell^2} R(\phi)$$

$$R_a = \frac{N}{A^2} \ell^4 \left[e^{B_a \frac{h^4}{N}} r_{0,a} \right]$$

m^{-3}

contact resistance

$$R_t = \frac{L^2}{N\ell^2} \left(r_{0,t} e^{\frac{B_t h^4}{N}} \right)$$

$B_a = B_t$ — from our data

$r_{0,a} = r_{0,t}$

$R(\phi)$ is contact resistance

$$R_a = \frac{N l^4}{A^2} e^{BAh/N} r_0$$

$$\sigma_a = \frac{h}{R_a A} = \frac{hA}{N l^4} e^{-BAh/N} \frac{1}{r_0}$$

$$R_t = \frac{L^2}{N l^2} e^{LWh/N} r_0$$

$$\sigma_t = \frac{L l^2}{R_t h W} = \frac{\cancel{L} N l^2}{\cancel{L^2} h W} e^{-LWh/N} \frac{1}{r_0}$$

$$\sigma_t = \frac{N l^2}{h(LW)} e^{-LWh/N} \frac{1}{r_0} \rightarrow \left[\frac{1}{\Omega m}\right]$$

$$\phi = \frac{N V_f}{A \cdot h} = \quad \Rightarrow \frac{N^2 l^6}{A^2 h^2} = \frac{\phi^2}{V_f^2} l^6$$

$$\frac{\sigma_t}{\sigma_a} = \frac{N l^2 \, N l^4}{h W L \, h A} = \frac{N^2 l^6}{h^2 A^2} = \frac{\phi^2}{V_f^2} l^6$$

$$\frac{\sigma_t}{\sigma_a} = \phi^2 \frac{l^6}{l^4 t^2} = \phi^2 \left(\frac{l}{t}\right)^2$$

$$\phi = \frac{N V_f}{A \cdot h}$$

$$V_f = l^2 t$$

$$\phi \simeq 0.3$$

$$\phi^2 \approx 0.1 \quad \frac{l}{t} \sim 10^3$$

$$\rightarrow \frac{\sigma_t}{\sigma_a} \sim 10^5$$ * Pre-factors